

Harish-Chandra Research Institute

Algebraic Topology

Assignment 1

M.Sc. Mathematics, Year II • Date of Submission: 26/08/2026

Problem 1

1. Verify that topological spaces and continuous maps form a category **Top**.
2. Let **Top**_{*} denote the category whose objects are pointed topological spaces (X, x_0) and whose morphisms

$$f: (X, x_0) \longrightarrow (Y, y_0)$$

are continuous maps satisfying $f(x_0) = y_0$. Verify that **Top**_{*} is a category.

Problem 2

Let X and Y be topological spaces. Recall that two continuous maps $f, g: X \rightarrow Y$ are homotopic if there is a continuous map

$$H: X \times I \longrightarrow Y$$

such that

$$H(x, 0) = f(x) \quad \text{and} \quad H(x, 1) = g(x)$$

for every $x \in X$.

1. Prove that homotopy is an equivalence relation on $\text{Hom}_{\mathbf{Top}}(X, Y)$.

If f is homotopic to g we denote it as $f \simeq g$.

2. Prove that if

$$f_0 \simeq f_1: X \rightarrow Y \quad \text{and} \quad g_0 \simeq g_1: Y \rightarrow Z,$$

then

$$g_0 \circ f_0 \simeq g_1 \circ f_1.$$

3. Deduce that one obtains a category **hTop**, called the homotopy category, whose objects are topological spaces and whose morphisms are homotopy classes of continuous maps.

Problem 3

Fix a topological space T .

1. Show that the assignment

$$X \longmapsto \text{Hom}_{\mathbf{Top}}(T, X)$$

defines a covariant functor

$$\text{Hom}_{\mathbf{Top}}(T, -): \mathbf{Top} \longrightarrow \mathbf{Set}.$$

2. Show that the assignment

$$X \longmapsto \operatorname{Hom}_{\mathbf{Top}}(X, T)$$

defines a contravariant functor from **Top** to **Set**.

Problem 4

Let $I = [0, 1]$. Consider the functors

$$F, G: \mathbf{Top} \longrightarrow \mathbf{Top}$$

defined by

$$F(X) = X \times I, \quad G(X) = X,$$

and

$$F(f) = f \times \operatorname{id}_I, \quad G(f) = f.$$

For each $t \in I$, define

$$\iota_{t,X}: X \longrightarrow X \times I, \quad \iota_{t,X}(x) = (x, t),$$

and define

$$p_X: X \times I \longrightarrow X, \quad p_X(x, s) = x.$$

1. Prove that

$$\iota_t = \{\iota_{t,X}\}_X$$

is a natural transformation $G \Rightarrow F$.

2. Prove that

$$p = \{p_X\}_X$$

is a natural transformation $F \Rightarrow G$.

3. Compute the composition $p \circ \iota_t$.

4. Explain why $\iota_t \circ p$ is not equal to the identity natural transformation of F . Prove, however, that it is *naturally homotopic* to the identity.

Problem 5

A continuous map $f: X \rightarrow Y$ is called a homotopy equivalence if there exists a continuous map $g: Y \rightarrow X$ such that

$$g \circ f \simeq \operatorname{id}_X \quad \text{and} \quad f \circ g \simeq \operatorname{id}_Y.$$

In this case X and Y are said to be homotopy equivalent.

1. Prove that homotopy equivalence is an equivalence relation on topological spaces.
2. Prove that a topological space X is contractible if and only if it is homotopy equivalent to a one-point space.
3. Prove that every convex subset of $\mathbb{R}^n$ is contractible.

Problem 6

Let X be a topological space and let $x_0, x_1, x_2 \in X$. If $\alpha: I \rightarrow X$ is a path from x_0 to x_1 , define the reverse path by

$$\bar{\alpha}(t) = \alpha(1 - t).$$

If α is a path from x_0 to x_1 and β is a path from x_1 to x_2 , define their concatenation by

$$(\alpha * \beta)(t) = \begin{cases} \alpha(2t), & 0 \leq t \leq \frac{1}{2}, \\ \beta(2t - 1), & \frac{1}{2} \leq t \leq 1. \end{cases}$$

1. Prove that $\alpha * \beta$ is continuous.
2. Show that concatenation is not strictly associative as an operation on parametrized paths by providing an explicit counterexample where $(\alpha * \beta) * \gamma \neq \alpha * (\beta * \gamma)$.
3. Construct an explicit homotopy relative to the endpoints between

$$(\alpha * \beta) * \gamma \quad \text{and} \quad \alpha * (\beta * \gamma).$$

Problem 7

Let (X, x_0) be a pointed topological space. Let $\pi_1(X, x_0)$ denote the set of homotopy classes relative to the endpoints of loops based at x_0 .

Prove carefully that the operation

$$[\alpha] \cdot [\beta] = [\alpha * \beta]$$

is well defined and makes $\pi_1(X, x_0)$ into a group.

Problem 8

Let

$$f: (X, x_0) \longrightarrow (Y, y_0)$$

be a pointed continuous map. Define

$$f_*: \pi_1(X, x_0) \longrightarrow \pi_1(Y, y_0), \quad f_*([\alpha]) = [f \circ \alpha].$$

1. Prove that f_* is well defined.
2. Prove that f_* is a group homomorphism.
3. Prove that

$$(\text{id}_X)_* = \text{id}_{\pi_1(X, x_0)}.$$

4. For pointed maps

$$(X, x_0) \xrightarrow{f} (Y, y_0) \xrightarrow{g} (Z, z_0),$$

prove that

$$(g \circ f)_* = g_* \circ f_*.$$

5. Conclude that the fundamental group defines a functor

$$\pi_1: \mathbf{Top}_* \longrightarrow \mathbf{Grp}.$$

Problem 9

Let $x_0, x_1 \in X$, and let λ be a path from x_0 to x_1 . Define

$$c_\lambda: \pi_1(X, x_0) \longrightarrow \pi_1(X, x_1)$$

by

$$c_\lambda([\alpha]) = [\bar{\lambda} * \alpha * \lambda].$$

1. Prove that c_λ is a well-defined group isomorphism. What is $(c_\lambda)^{-1}$?
2. Suppose that μ is another path from x_0 to x_1 . Show that the composition $c_\lambda \circ (c_\mu)^{-1}$ is given by an inner automorphism of $\pi_1(X, x_1)$. Explicitly identify the element of $\pi_1(X, x_1)$ defining this conjugation.

Problem 10

1. Let

$$f, g: (X, x_0) \longrightarrow (Y, y_0)$$

be pointed maps that are homotopic through a pointed homotopy. Thus, suppose that there exists a homotopy

$$H: X \times I \longrightarrow Y$$

such that

$$H(x, 0) = f(x), \quad H(x, 1) = g(x), \quad H(x_0, t) = y_0$$

for all $x \in X$ and $t \in I$. Prove that

$$f_* = g_*: \pi_1(X, x_0) \longrightarrow \pi_1(Y, y_0).$$

2. Deduce that a pointed homotopy equivalence induces an isomorphism of fundamental groups.
3. Prove that

$$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0).$$

4. Compute

$$\pi_1(X \times \mathbb{R}^n, (x_0, 0))$$

in terms of $\pi_1(X, x_0)$.